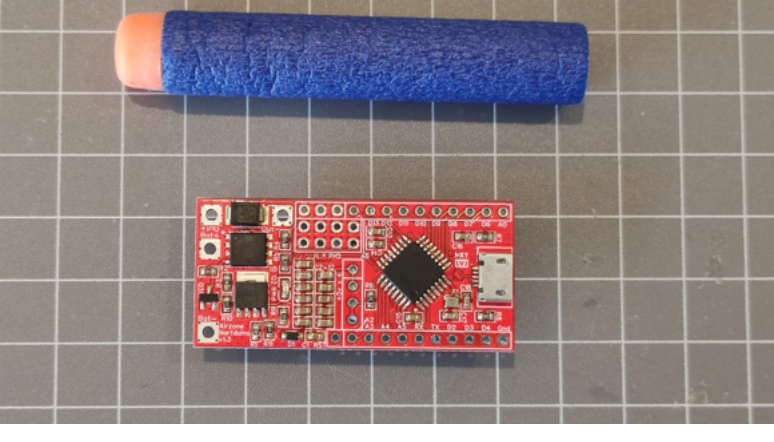


Blasters By Airzone

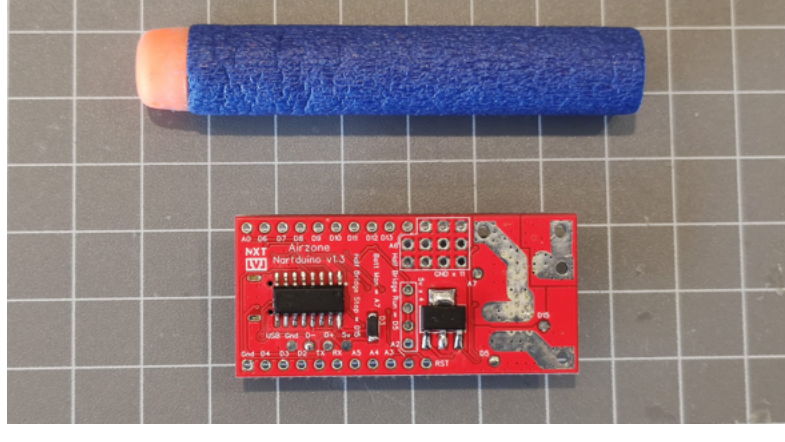
Narduino

Narduino is a microcontroller designed specifically for high performance, software controlled foam dart blasters. It essentially has all the capability of an Arduino Nano, but integrates the following additional features:

- Power supply tested on 2s to 4s LiPO packs. It should be fine on 5s, but this is untested.
- Integrated Half-H-Bridge to give DC motor drive and braking directly on-board. Rated for approx 20a continuous, with bursts of 80a. Exposed traces can be built up with solder or bus bars to increase current rating. You can also attach a solenoid instead of a motor
- Integrated battery voltage monitor
- ATMega328p microcontroller. All pins are broken out for use, with the exception of D5, D15, and A7 – which are for the Half-H-Bridge and battery monitor.
- USB Serial bridge. Connect directly to your PC by USB cable to program. Break-out pads are included for USB so you can run a remote USB port, if you like.
- 12 available ground pins, and 5 available 5v pins.



Front side of Narduino with date for scale



Rear of Narduino

Half-H-Bridge

Pins D5 and D15 are consumed by the half-bridge. Although this is designed for a motor, a solenoid can be substituted in without any dramas. Obviously motor braking does not apply to a solenoid.

D5 is connected to the drive (low side) mosfet. Bringing D5 high will start the motor. Bringing D5 low will free-wheel the motor. D5 is PWM enabled, and offers speed control.

D15 is connected to the brake (high side) mosfet. Bringing D15 high will short the motor terminals through the mosfet, and provide braking. Bringing D15 low will keep the brake off.

When using a solenoid instead of a motor, wire it the same way. D5 will control the pulse, and D15 should be kept low at all times.

Caution:

Having both the drive and brake mosfets high at the same time will lead to shoot-through and subsequent detonation of the Narduino, potentially any connected computer, any other electronics connected to the board, and also potentially your LiPO battery. Mosfet gates act like capacitors. It will take some time from when the gate is brought low to the mosfet actually turning off. Not much time, but still some. You need to let one gate discharge before activating the other. This is dead-time generation.

Battery Voltage Monitor

Voltage monitor is connected to pin A7 and is set up as a simple resistor divider between 10k and 47k resistors. Resistors are 1% accuracy and therefore a calibration factor might be required

Example formula:

BatteryCurrentVoltage = (((float)Sample * 5.0) / 1024.0 * 5.7) + BATTERY_CALFACTOR;

Sample - the value read from an analogRead()
BATTERY_CAL_FACTOR = a constant/proportional adjustment. You will need to measure with a multimeter and compare to the output to determine what this is.

Power Supply

A SOT223 LDO provides power from the LiPO through a general purpose diode (to act as a non-return for stable MCU operation). 242uF in surface mount caps provide some buffer against voltage sag and results in stable operation.

Maximum theoretical output is 1a, however without strong cooling, 200-300ma is more reasonable. This is easily enough for Narduino, an OLED, etc.

Reliable operation is not guaranteed on a discharged 2s LiPO due to the voltage loss through the diode and LDO.

Version 1.6 and higher uses a high-efficiency switch-mode power supply instead of an LDO. You will get more sustained power through this upgraded system.

Pin Functions

A	D	Function	PCINT	PWM	Chip Pin
A0	D04	+	Y		PD0
D6	SC0A	AIN0	22	Y	PD6
D7	AIN1	23	N		PD7
D8	CP0A	CL00	0	N	PB0
D9	OC0A	1	Y		PB1
D10	OC1B	2	Y		PB2
D11	MOSI	OC2	3	Y	PB3
D12	MISO	4	N		PB4
D13	SCK	5	N		PB5
A6			N		AD06
		GND			
		GND			
		BRIDGE GROUND			

BRIDGE +

BATTERY +

BRIDGE GROUND

BATTERY GROUND

Chip Pin	PWM	PCINT	Function	D	A
PD4	N	20	SCX	T0	D4
PD3	Y	19	SC2B	INT1	D3
PD2	N	18	INT0		D2
PD1	N	17	TXD		D1
PD0	N	16	RXD		D0
PC5	N	13	SCL	D15	A5
PC4	N	12	S0A	D16	A4
PC3	N	11		D17	A3
PC2	N	10		D18	A2
PC6	N	14	RESET		

Board Functions (Pads On Rear)

Chip Pin	PWM	PCINT	Function	D	A
PD5	Y	21	HB-DRIVE	D5	
PC1	N	9	HB-STOP	D19	A1
AD07	N		BATT-MON		A7

BRIDGE +

BATTERY +

BRIDGE GROUND

BATTERY GROUND

Digital Pin

Analog Pin

Port Pin

Pin Function

Serial Pin

Control Pin

Interrupt Pin

ESC Pin

5v Power

GND

Narduino Pinout

Pinout Diagram - [Open in New Window](#)

Pads on rear are provided for D5, D15, and A7 to allow for monitoring and diagnostics.

Pads on rear are provided for USB +5v, GND, D+, D- to allow for a remote USB plug to be wired in.

Battery + and Bridge + are connected internally. Connect one to your LiPO positive terminal and the other to your motor / solenoid.

Battery Ground connects to your LiPO negative terminal.

Bridge Ground connects to your motor / solenoid.

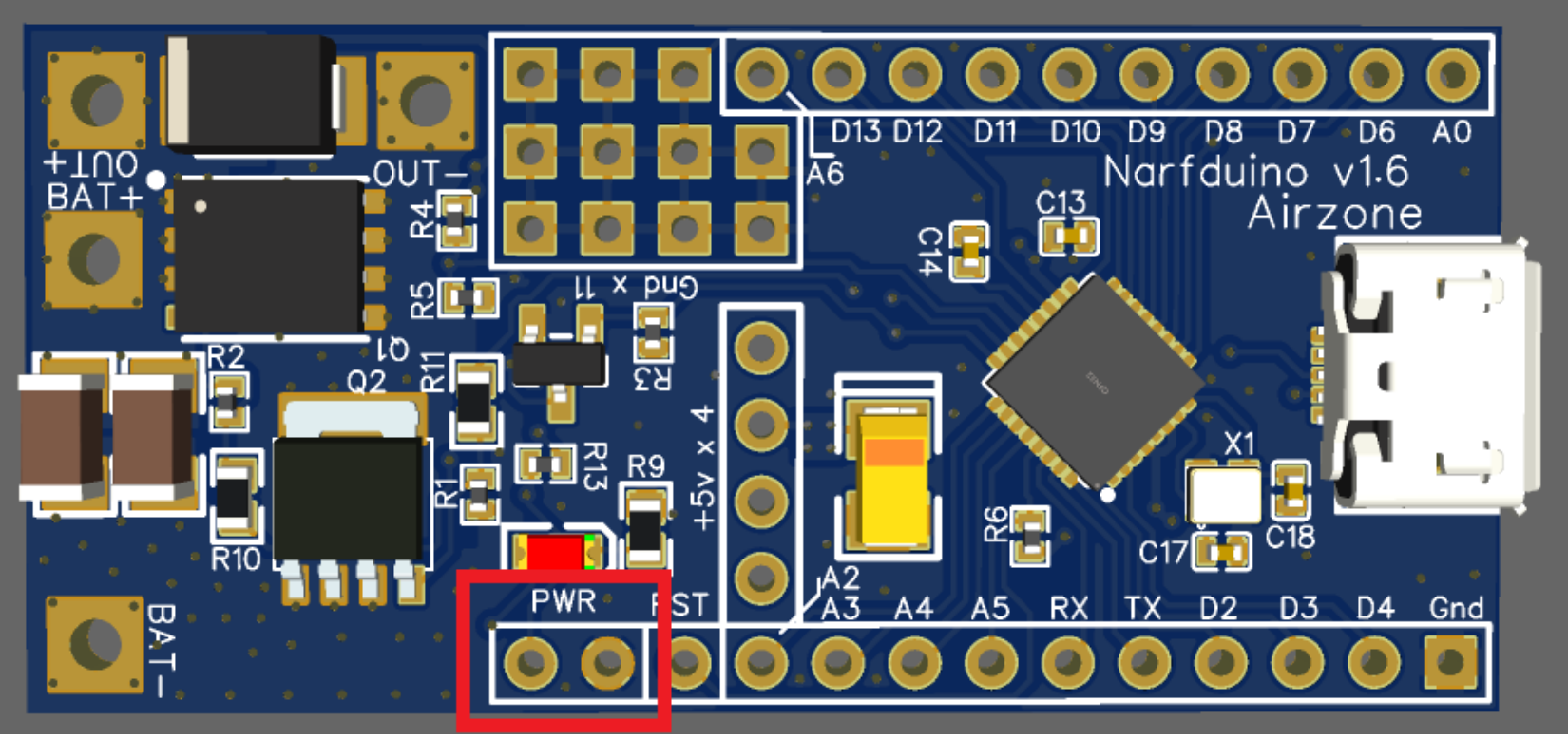
Caution

Bridge Ground and GND pins must not be soldered together. Pay attention!!
Do NOT attempt to power accessories from USB +5v pad. This is for input only!

v1.6+ Addendum

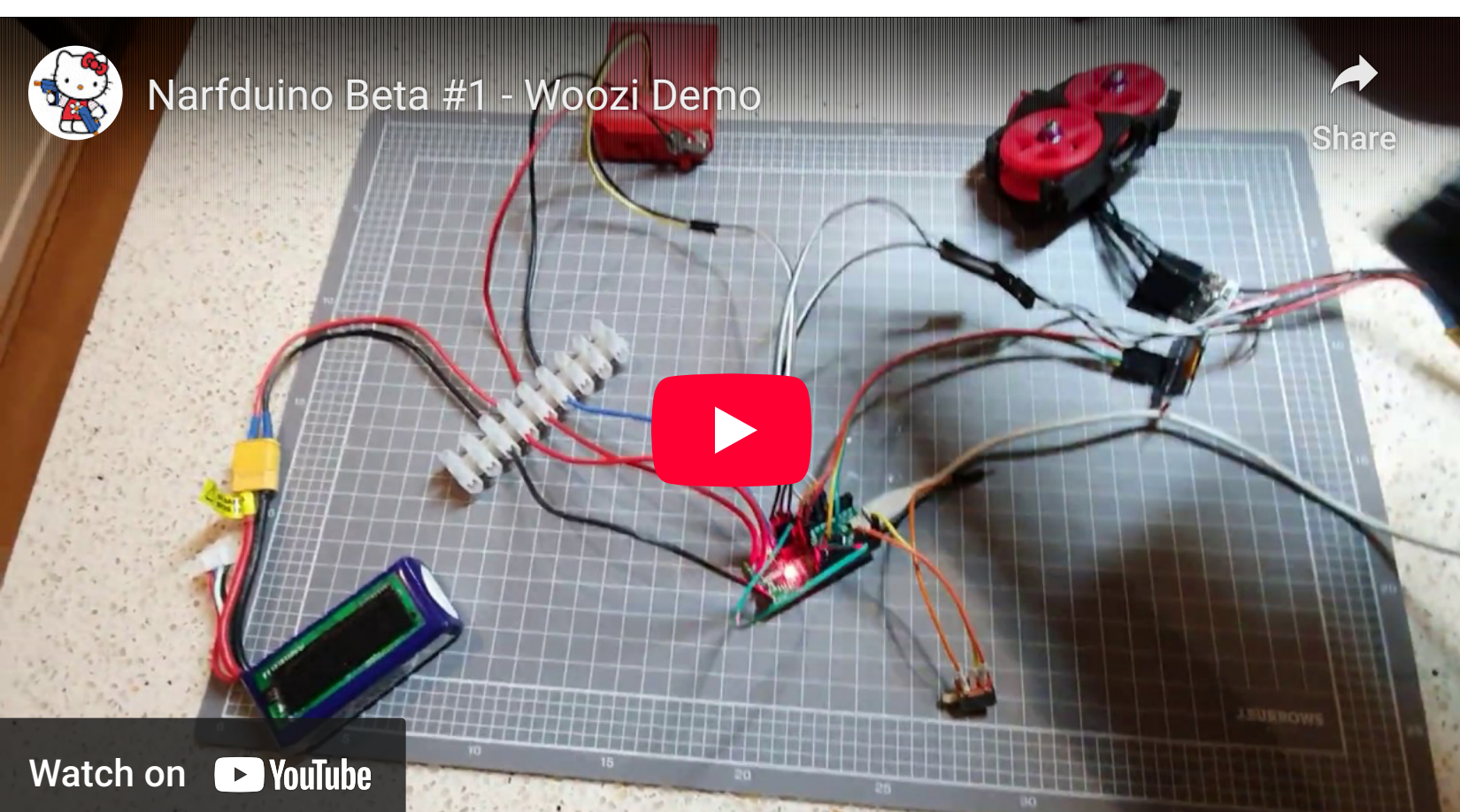
Starting from v1.6, the power supply has been changed to a more efficient switch-mode power supply (SMPS). This means it can handle higher currents for longer without getting hot. In addition, a pair of 'Power' pins have been added to act as a power switch. If you do not use these pins, the Narduino will behave like normal and will be always-on. If you connect a switch, the Open position will be On, and the Closed position will be Off. The +5v pins will be turned off as well. The current draw across these pins is miniscule, so any switch will do.

Warning: The Off position puts the power supply in stand-by mode, but you should not use this as a substitute for disconnecting your battery. It will still run your battery down over the course of about 2 weeks or so.

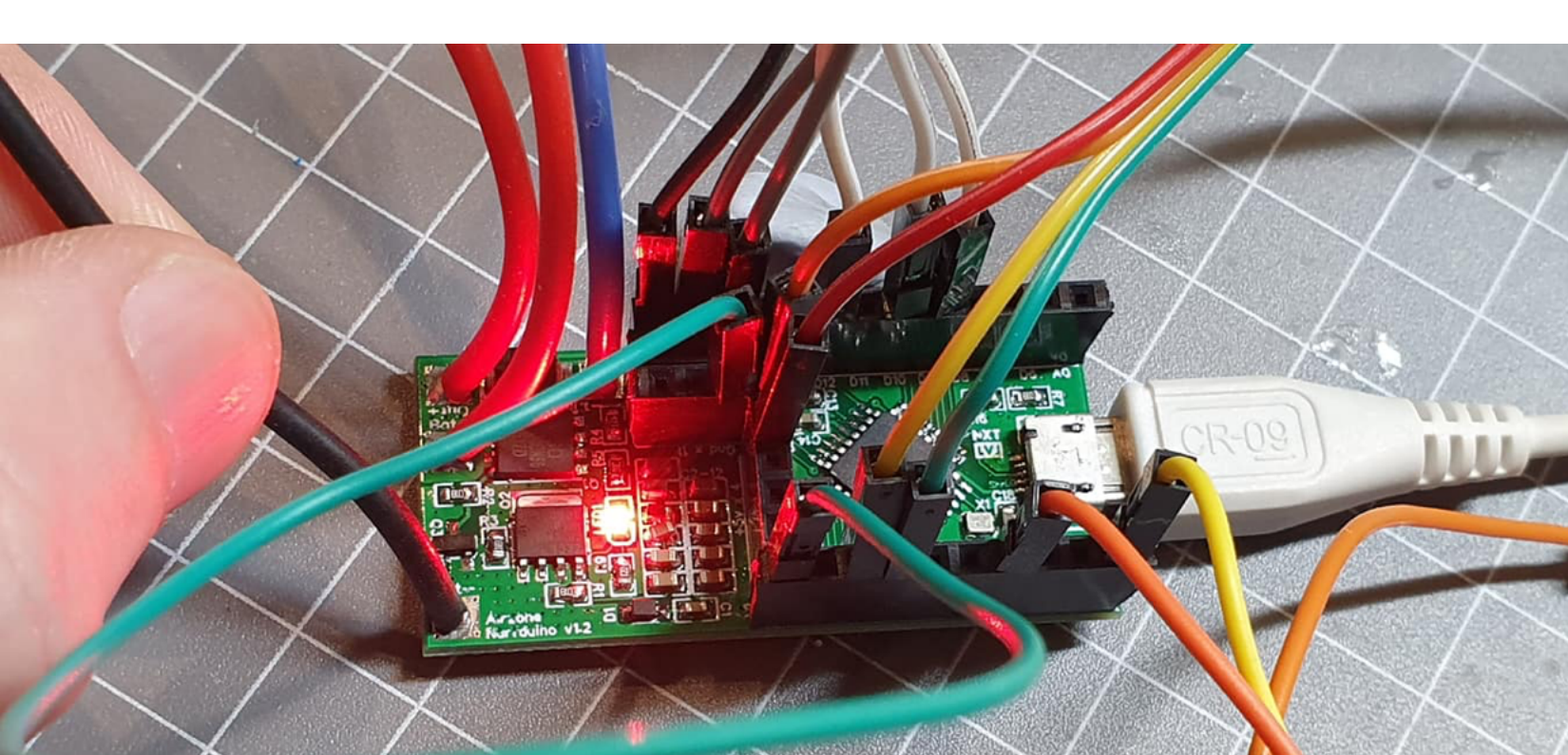


Solder your power switch to these pins.

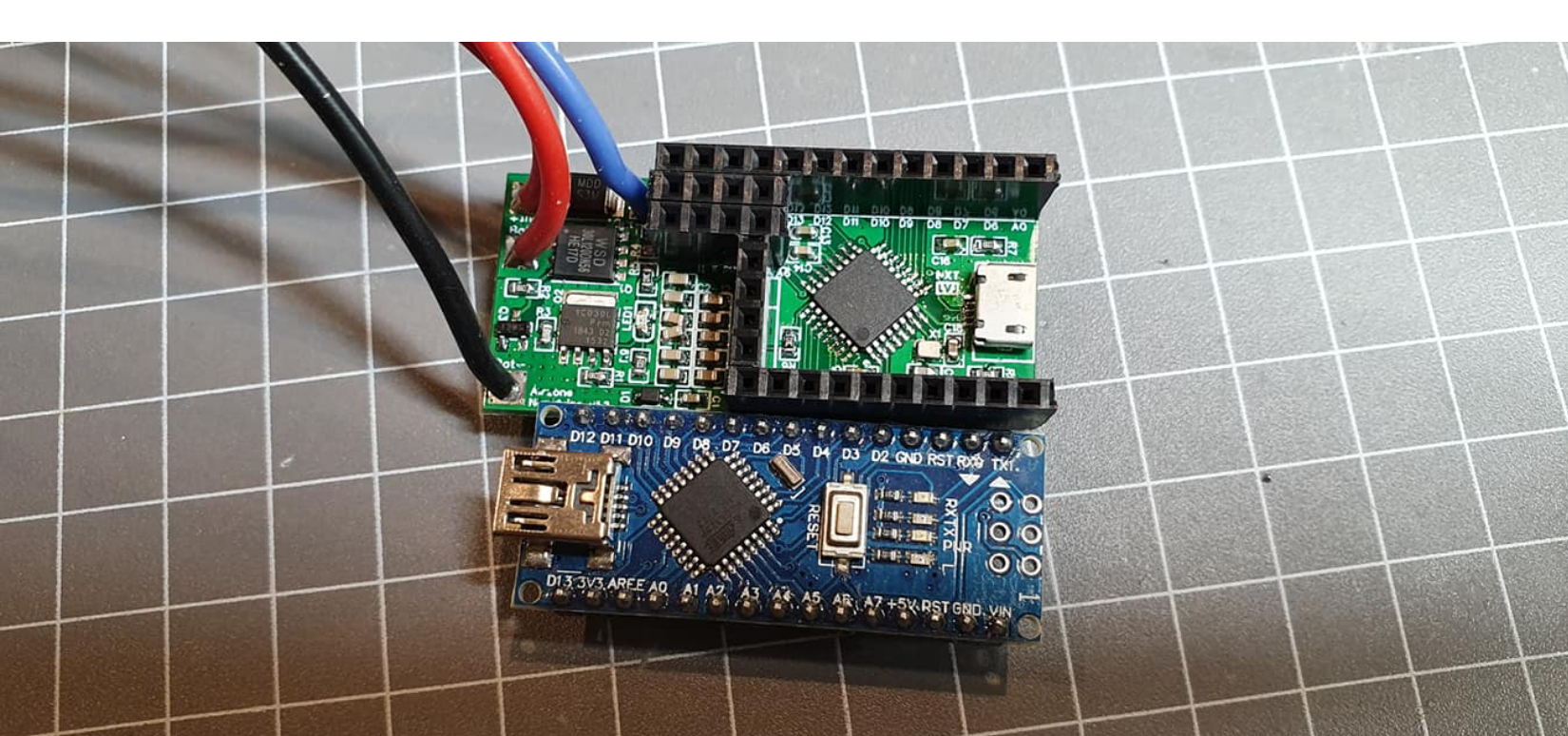
Media



Demonstration of Narduino using a test bench setup



Connected to an OLED screen, a Woozi pusher, and brushless motors.



Prototype comparison to Arduino Nano

Errata

v1.2 – (Beta units) There is no backfeed protection on the USB port. It is strongly advised to NOT connect your PC to the Narduino at the same time as a LiPO.

v1.3 – Some individual Narduino boards are populated with a different battery sense resistor network. This doesn't affect functionality or reliability, but will require a different formula to be used. You will know if you have an affected board if the default formula produces wildly incorrect results (e.g. 1.3v instead of 11v):

BatteryCurrentVoltage = (((float)Sample * 5.0) / 1024.0 * 48.0) + BATTERY_CALFACTOR;

Warning

You are responsible for your own half-bridge dead-time generation and shoot-through protection. Failure to provide appropriate dead-time will cause fireworks. A control library is available to download and install into Arduino that will take care of this for you.

Failure to provide jam detection or manual abort capability on pushers and flywheels may lead to a stalled motor burning up it's windings. Depending on the motor, this may damage the Narduino

It is not recommended to connect your Narduino to your computer and LiPO battery at the same time. Although there is back-feed protection circuitry built in, I won't be responsible if something happens and you lose a USB port and/or motherboard.

This product is designed for high-performance blasters. Please exercise due caution when modifying and using modified blasters.

Warranty

All boards are tested prior to shipping. If for some reason it's dead on arrival, please contact me.

Once soldered to, all warranties and return rights are voided.

This is an advanced modification product and is not sold as a drop-in. If you are not sure how to go about building a blaster, contact me.

Buy Narduino



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